

Mini Review

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Development, spread and persistence of antibiotic resistance genes (ARGs) in the soil microbiomes through co-selection

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Abstract: Bacterial pathogens resistant to multiple antibiotics are emergent threat to the public health which may evolve in the environment due to the co-selection of antibiotic resistance, driven by poly aromatic hydrocarbons (PAHs) and/or heavy metal contaminations. The co-selection of antibiotic resistance (AMR) evolves through the co-resistance or cross-resistance, or co-regulatory mechanisms, present in bacteria. The persistent toxic contaminants impose widespread pressure in both clinical and environmental setting, and may potentially cause the maintenance and spread of antibiotic resistance genes (ARGs). In the past few years, due to exponential increase of AMR, numerous drugs are now no longer effective to treat infectious diseases, especially in cases of bacterial infections. In this mini-review, we have described the role of co-resistance and cross-resistance as main sources for co-selection of ARGs; while other co-regulatory mechanisms are also involved with cross-resistance that regulates multiple ARGs. However, co-factors also support selections, which results in development and evolution of ARGs in absence of antibiotic pressure. Efflux pumps present on the same mobile genetic elements, possibly due to the function of Class 1 integrons (*Int1*), may increase the presence of ARGs into the environment, which further is promptly changed as per environmental conditions. This review also signifies that mutation plays important role in the expansion of ARGs due to presence of diverse types of anthropogenic pollutants, which results in overexpression of efflux pump with higher bacterial fitness cost; and these situations result in acquisition of resistant genes. The

future aspects of co-selection with involvement of systems biology, synthetic biology and gene network approaches have also been discussed.

Keywords: antibiotics; antibiotic resistance genes; co-selection; cross resistance; heavy metal; poly aromatic hydrocarbons; soil microbiome.

Introduction

The development of antibiotic-resistance in bacteria due to its exposure to different antibiotics, heavy metals and poly aromatic hydrocarbons (PAHs) has promptly increased in the past few years. Because of non-availability of therapeutic alternatives to treat multidrug resistance (MDR) bacteria [1, 2], use of antibiotics is increasing, and hence resistance for antibiotic is also increasing. The mode of resistance in bacteria such as *Stenotrophomonas maltophilia*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae* HS11286, *Acinetobacter baumannii*, *Escherichia coli* derivative QepA2, etc. varies prominently and includes antibiotic detoxification, target substitution and protection and block the antibiotic accumulation [3]. However continuous use of antibiotics leads to mutations into the bacteria and gradual build-up of MDR genes in their genomes, which lead to development of resistant phenotypes [4, 5] that have capacity to survive under the higher pressure of antibiotics [6]. Further, it may be hypothesize that due to variation in usage of multiple types of clinical and non-clinical settings might plays an important role in the persistence of MDR bacteria after development of co-selection which causes proliferation of co-resistance into the environment [7].

Co-selection may be defined as “development and persistence of antibiotic resistance in microbial population, even when the antibiotics are not present”, or present up to sub-lethal dose only” [8]. In fact, mutations are mostly not involved in this process, instead horizontal gene transfer plays important role and there were confirmations for transfers of resistance genes among diverse

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group of organisms i.e., *S. maltophilia*, *Clostridium perfringens*, *Burkholderia cepacia*, *Enterococcus faecalis*, *Streptococcus pneumoniae*, *Ochrobactrum intermedium*, and *Bacteroides* [9]. There are mainly two driving force for co-selection mechanisms: co-resistance and cross resistance [10]. However, direct co-selection of antibiotic resistant genes (ARGs) in the bacterial population such as *Enterococcus faecium*, *Aeromonas salmonicida* subsp. *Salmonicida*, and *E. coli* can be attributed to different pollutants, which may be antibiotics itself, heavy metals, or even poly aromatic hydrocarbons (PAHs) [11]. Since, these co-factors supports the selection, and hence evolution of resistance, despite being not present up to lethal concentration to exert pressure by these compounds, and therefore co-selection intensely increases the density of the antibiotic resistance from human to environment as well as from environment to human [9]. Further these co-factors became more predominant in the environment where discharge of the contaminants is rigorous and continuous in time [11]. Therefore, it may be speculated that, the pressure caused by the co-factors adds the indirect co-selection of ARGs, particularly in the environments where multiple contaminants are presents.

Like antibiotics, heavy metals are known to exhibit the wide range of antibacterial activity. In earlier days when therapeutic alternatives were limited, the antimicrobial metal compounds like Cu, Ag, K, Mg, Zn, Bi, etc. were used in the health care settings. Some of them are listed by the WHO (accessed on 17th Feb 2019. <https://www.who.int/medicines/publications/essentialmedicines/en/>). It has been reported that Bismuth (Bi) may be used as drug, prescribed and given with antibiotics to treat infections associated with *Helicobacter pylori* [12] and can also be used for clarithromycin- and metronidazole resistant infections [13, 14]. There are many evidences to support that; the antibiotic resistance may get derived from exposure of heavy metals such as Zn and Cu. This may happen because selection of heavy metal mobile genetic elements harbours both metal and antibiotic resistant genes (ARGs) [15–20] (Figure 1). Further, some of the antibiotics make complexes with heavy metals and by doing so, enhance or reduce their activities [21, 22].

In addition to this, PAH pollutants are generally coming from vehicle emission, tarmac, paint, tires, oil refineries, crude oil spilling, from automobiles and road makings [23] and these PAHs also have the ability to enrich the ARGs in soil [24, 25] and coastal waters as well [26] (Figure 1). ARGs have been reported to be present 15 times higher in PAH contaminated soils [22] and are considered

as the emerging environmental contaminants (Figure 1) [10, 27].

Environmental influence of ARGs by co-selection in presence of heavy metals, antibiotics and PAHs

Co-selection of antibiotic with heavy metal, or PAH resistance promote ARGs in microbial community in spite the absence of any selective pressure. It's true that cross-resistance and co-resistance are the main drivers for co-selection of ARGs but some other mechanisms such as regulatory mechanisms are also involved in cross-resistance that consist of the spread of multiple resistance genes which are regulated by a single regulatory gene. Essentially, co-selection may be attributed to the following three main phenomenon's (Figure 2).

- A. Cross-resistance: It occurs when one resistance genes showing resistance to too many heavy metal or antibiotics or PAHs and happens via efflux pump which provides resistance to different compounds including antibiotic, PAHs and heavy metals simultaneously [17] In fact, MexGHI-Opmd efflux pump in *P. aeruginosa* has been attributed for the resistance to clavulanic acid, vanadium and ticarcillin. Additionally, MexGHI-Opmd plays an important role in quorum sensing (QS) response in *P. aeruginosa* in a secondary mode and exports the toxic metabolite [28].
- B. Co-resistance: Physical presence of different resistance genes are on the same mobile element like transposons or plasmids [2, 23] may lead to development of co-resistance. For example, the operon for the resistance against Cu and Ag, and a beta lactamase gene (*bla_{CTX-M}*) are usually present on same plasmids. These plasmids also carry the genes for quaternary ammonium compounds (QACs) [29].
- C. Co-regulatory mechanism: It regulates the multiple resistance genes which are resistant to different toxic compounds like heavy metals or antibiotics [15]. For example, two-component regulator of *P. aeruginosa* i.e., CzcRS increases the expression of the *czcCBA* efflux pump system which are responsible for resistance to Cd, Zn and Co, and further responsible for decreasing the expression of OprD porins which are leading to increasing resistance to antibiotics like imipenem [30].

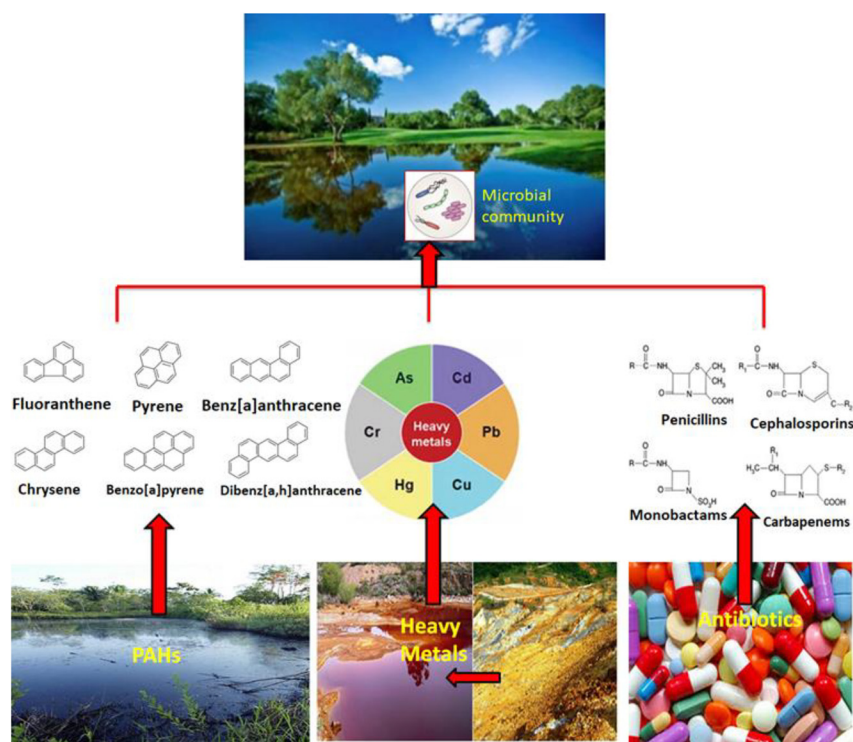


Figure 1: Overview of microbial communities exists in the environment are subject to diverse pollutants: PAHs, heavy metals, and antibiotics, etc. These conditions encouraging a co-selection process which is indirectly selecting for ARGs into the normal environment [Abbreviations: PAHs – poly aromatic hydrocarbons; ARGs – antibiotic resistance genes; As – arsenic; Cd – cadmium; Pb – lead; Cu – copper; Hg – Mercury; Cr – chromium].

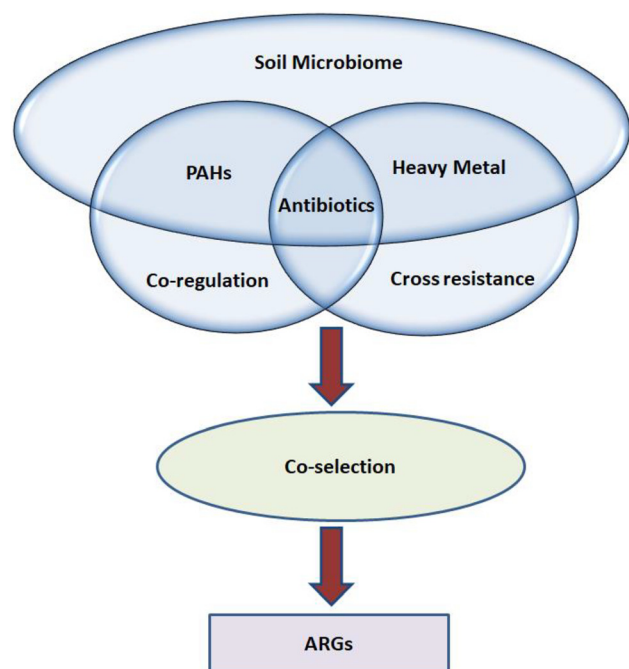


Figure 2: PAHs, Heavy Metal, and antibiotic Co-selection in the soil microbiomes. When common resistance machinery devoted to resistance to antibiotics, heavy metal and/or PAHs; a phenomenon termed as cross-resistance. Co-regulation or co-expression happening when the expression of resistance machinery antibiotics, heavy metals, and PAHs are controlled by a common regulator at the same time in the environment [Abbreviations: PAHs – poly aromatic hydrocarbons; ARGs – antibiotic resistance genes].

Co-occurrence of heavy metal resistance and antibiotic cross resistance

In bacterial systems, efflux pumps are the main molecular devices that provide resistance against heavy metals [20]. It is the fact that efflux pumps are complex machines which pump molecules from cytoplasm through the membranes and cell wall (Figures 3, 4). This generic mechanism expels the different products of bacterial metabolisms, but minimal levels of efflux are not enough for the resistance against heavy metals. However, over-expression or mutations in these efflux pumps can expand the range of the molecules extruded by the efflux system and results in leading to the resistance to many compounds such as antibiotics, PAHs, heavy metals and biocides (Figure 3) [23, 31].

The heavy metal resistance mechanisms usually have metal ion-specific efflux proteins, and other protein complexes that facilitate their efflux, and/or enzymes that convert it to less toxic form. These are coded by heavy metal resistance genes, which are regulated by metal-ion specific response regulator gene [10]. Studies have demonstrated that contaminations of heavy metals like Cu, Cd, Pb and As are positively correlated to fate and distribution of ARGs, which suggests that these metals impose

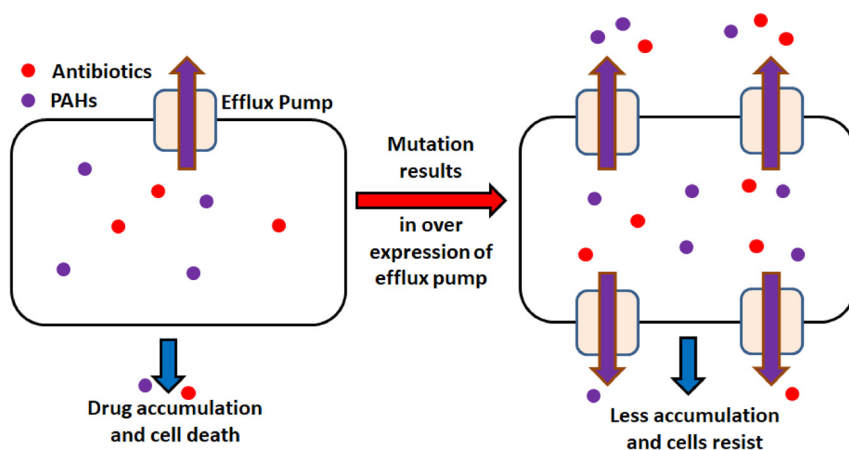


Figure 3: Efflux pump mechanisms of cross-resistance, and co-regulation/co-expression of heavy metal, antibiotics, and poly aromatic hydrocarbonss (PAHs). Altered target or mutation is happing due exposure to heavy metal, PAHs or antibiotics etc., and which leads to the efflux pump over expression.

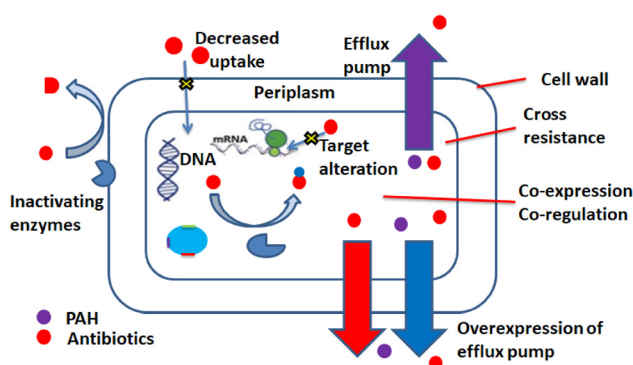


Figure 4: Effect of Heavy metal, Antibiotic and poly aromatic hydrocarbonss (PAHs) on efflux pumps are causing several mutations or lesions, which leads to overexpression and bacteria resisting themselves to heavy metal, antibiotics or PAHs.

stress which results in co-selection of ARGs (Figure 4) [18]. Further study is also showing the presence of the *sulI*, *tetM*, *tetA*, *qnrA* and *qepA* genes had a progressive association with heavy metals like Cu, Hg and Zn [32]. *sulI* has role in sulfonamide resistance and code for dihydropteroate synthase, while *tetA* and *tetM* code for tetracycline resistance. *qnrA* encodes for quinolone resistance, and *qepA* codes for fluoroquinolone specific major facilitator subfamily (MFS)-type efflux pump.

However, the link between heavy metal and ARGs is quite common, as both the resistance genes are present on the same genetic elements [1, 18]. Yet higher concentrations of heavy metal in the environments may apply the co-selection pressure that induces antibiotic-resistance by genetic couplings [17, 33], especially in the soil environment where these two class of contaminants are present [15]. It is well known that antibiotic resistant bacteria's (ARBs) of clinical significance are present in large numbers in the environments contaminated with heavy metal [20]. Therefore, it is significant to mention that abundance of

heavy metals and antibiotic resistance genes can promptly change as per environmental conditions or pressure, as these genes present in mobile genetic elements specially class 1 integrons (*Int1*) which are present in several bacterial species.

Co-occurrence of antibiotic and PAHs cross resistance

PAHs present in the environment significantly influencing the microbial communities [10]. The most predominant genera in PAH contaminated soils microbiome is proteobacteria, however significant abundances of other bacteria have also been observed [16, 17]. Hence, it may be presumed that PAH-contaminated soil microbiomes are enriching ARGs which are then shared among pathogenic bacteria by horizontal or vertical transmission mechanisms. Also, it has been reported that ARGs are upregulated into PAH contaminated areas [10, 16, 26, 34]. PAH are hydrophobic in nature, and therefore it forms strong associations with non-polar hydrophobic moieties in soil after getting in it [35]. Often, due to this, it becomes difficult to assess the actual level of PAH contamination, and therefore its impact on the progress of ARGs in the soil microbiomes.

The processes that enrich the ARGs in PAHs contaminated soil is not well understood but there is substantial proof that the bacterial efflux pumps, which pump out the pollutants such as PAH are homologous to multidrug efflux pumps which are encoded by ARGs (Figures 3, 4) [16, 23, 31]. *Int1* is known to integrate these genes to the same mobile genetic elements, and it's interesting that *Int1* was present in higher abundance in PAH contaminated soil [26]. Proteobacteria had been the dominant phylum of bacteria in PAH contaminated soils, and were found to be

enriched with ARGs [25, 26]. Though, these ARGs were predominantly for efflux pump encoding genes related to aromatic antibiotics. Further, these were not plasmid borne, but present on the chromosomes of bacteria, that could tolerate PAH. This indicated that the transfer of these genes was not mediated by plasmids [16, 19].

Although, there are some reports that contamination of PAH in soil reduces the horizontal gene transfer (HGT) of ARGs. Further, the frequency of vertical transfer of ARGs was also significantly low for smaller PAHs such as phenanthrene and also less significant in pyrene, probably due to their non-covalent binding with the plasmid, but there were no significant effect on vertical transfer rates for benzo[a]pyrene [25]. However, the location of efflux pump genes on chromosome suggests that the ARG enrichments mechanisms in PAH contaminated soils are different from heavy-metal contaminated soils [16, 36]. But, it may be assumed that in case, where HGT is interrupted in PAH contaminated soil, the copy number of ARGs will still increase with the progression of ARGs harboring bacteria.

Significance of co-resistance in co-selection of ARGs

There are several reports of co-existence of antibiotic-resistance and heavy-metal resistance genes [15, 17, 19], but there is huge lack of knowledge about co-existence between antibiotic-resistance and PAHs-related genes. In fact, there is genetic linkage between resistance for Mercury and ARGs, on plasmids which were co-transferred by conjugative process among members of *Enterobacteriaceae* and other bacteria [37]. The occurrence of multiple pollutants such as heavy metals, PAH, etc. act as co-factors in increasing the existence of ARGs however the evidence for these factors to co-select the ARGs in contaminated soil is still not known [16, 38]. Farias et al. identified genetic determinants such as *ACR3* and *arsB* genes from the amplicons of the total DNA. These resistant determinants showed Cu, Cr, Cd, Zn and Co resistance and also resistance to β -lactam as well as fluoroquinolones group of antibiotics. Among them the ARGs and heavy metal resistance genes were present in the same mobile genetic element [38]. Chen et al. reported that in PAH contaminated soils; ARGs were linked to more than 10 multiple antibiotic groups. Among them, 50% of the ARGs linked with chloroamphenicols, fluoroquinolones and polypeptides etc. However efflux pump-encoding ARGs linked with the fluoroquinolones and acriflavine, which were projected around 70% of entire ARGs in PAHs contaminated soils, and analysis of subtypes of ARGs advised that in contaminated soil PAH act as

substrate for *acrA*, *acrB*, *bacA*, *mexB*, *tolC* and *amrB* efflux transporters [16]. However in another study Yazdankhah et al. described that potentially toxic metals (PTMs) Hg, Cd, As, Ni, Pb and Co, could be present in a smaller amount in to the fertilizers, and might be act as a major source for expansion of antibiotic resistance in bacteria due to the interface among PTMs, antibiotics and disinfectants and bacteria [17]. In addition to this, plasmids of *Salmonella abortus equi* have elements for co-resistance for ampicillin with many metals like Cd, Hg, As and Cr [39].

ARGs, heavy metal or PAH resistance genes are most frequently isolated from the contaminated soils [19] which are inhabited by domestic animals and humans [25]. Hence, the microbial population which interacts also included the bacteria associated with animals and humans, along with soil indigenous flora. Such interactions are significant in co-occurrence of antibiotic-resistant genes because microbial populations associated with domestic animals and humans have to thrive through multiple antibiotics, and therefore, pressure associated with it, while soil isolates are exposed to PAH and heavy metals [40]. Though, this needs to be experimented in detail before reaching any substantial conclusion.

Further, *mcr-1* genes had been reported from Cu-resistant salmonellae and therefore, copper has been discouraged for any veterinary applications [41]. In addition to these, the cadmium and zinc resistance genes also shown to be irregular for co-occurrence with antibiotics like macrolides and aminoglycosides [10, 25]. Therefore, not only copper but also other heavy metals possibly co-select for ARGs in bacteria. Hence, unusual practice to use these metals should be avoided in common practice with antibiotics to control bacterial community to become co-resistance.

Bacterial fitness cost is an important aspect for resistance expansion

Bacterial fitness cost may be defined as endurance of an organism in heavy metals, PAHs or antimicrobial contaminated environment [42]. The fitness cost is an important parameter in the progression of co-resistance as it is directly associated with the competence of bacteria in the selective environment of either antibiotic or heavy metal or PAHs [43]. Notably, it contribute significantly for development of resistance forces by forming selective pressure for individual heavy metal or antibiotic or PAHs, when bacteria interact in environment.

Although fitness of resistance is flexible, and the factors responsible for it are still unclear, still there is substantial evidence which suggest that, chromosome mediated resistance through mutation activities provide greater fitness cost to organism as compared to genes acquired through plasmids [44] and hence this is the main basis of why genetic events at chromosome act as major source and evolution of ARGs in the environment [42]. Santos-Lopez et al. found variations in the susceptibilities among 13 different antibiotics depend upon the growth and mode of prior selection, when tested the minimum inhibitory concentration of 23 antibiotics in the planktonic environments of evolved population's. Authors further recorded that, planktonic generated inhabitants showed cross resistance to 3rd generation cephalosporins such as cefpodoximece, and ceftazidime, and was genetically linked with *gyrA*, *adeN*, *pgpB* or *adeS* mutations [44]. Vogwill and MacLean also confirmed that, chromosomal resistance mutation convey a greater fitness cost than the resistance through plasmid. Authors also suggested that, the selection of ARGs in clinical settings leads to the fusions of multiple mutations which pay at least marginal epistatic rate of antibiotic resistance. However, once resistance progresses through mutation in genes; the epistatic interfaces in between the resistance modifications have significance to affect the complete fitness cost related with resistance [45]. Though, it has also been observed that the rate of plasmid uptake gets significantly enhanced in presence of selective pressure, and resistance, which indicates that fitness cost can regulate the evolution of ARBs through plasmids [42, 44, 45].

Involvement of systems biology, synthetic biology and gene network approaches to understand antibiotic resistance

At present, there is a requirement for systems biology, gene network and synthetic biology approaches to differentiate the relationship between genes, pathways and proteins in bacterial metabolisms that will increase our knowledge for: in what way bacteria react, and protect themselves once they exposed to different concentration of antibiotics, PAHs or heavy metals? However the effects of the fundamental rationality of genetic networks are hard to infer over the experimental approaches only, however effective methods can do it in several ways, which comprises the amalgamation of novel experimentations and computational modeling methods. To report this issue, many

authors done several excellent computational experiments that enable us to construct the models of bacterial metabolic regulatory gene networks, via gene regulation quantification methods. And these developments are supposed to be the commencement to fix the platform for an integrated explanation of the bacterial metabolic regulatory routes that intricate in elementary cellular functions of the bacteria present in the soil microbiome [46–52].

Using the system biology and OMICS expertise, Martin-Galiano and McConnell, identified the epitope targets to develop the monoclonal antibodies against the ARBs. These monoclonal antibodies have shown many features usually required for the treatment of bacterial infections [53]. Camp et al. suggested that the high throughput sequencing approaches should be used to explore the bacterial pathogenicity that allowed the progression of huge quantities of data using bioinformatics investigation to characterize the novel ARGs and ARBs [54]. However, to understand and recognize the effect of metabolic variations in deliberation for antibiotic resistance, Zampieri et al. used a genome-scale model of *E. coli* metabolism; and established innovative constraint-based metabolic regulatory approaches [55]. While, Rodrigo et al. noticed that, input function arises as a tool to separate the basis of many significant genes for the *mar* phenotypes [56]. Such unique interventions are required to be explored to elucidate the interactions between PAH and ARGs.

Future perspective

The threats of antibiotic resistance and therefore the mechanisms associated with it find attention due to high frequency of MDR phenotypes with possible expansion of co-resistance in the environment. However, the maintenance of ARGs via a common genetic mechanism reduces the bacterial fitness cost, but increases the chance of spread and survival of co-resistant phenotypes. Hence, it is absolutely needed to recognize the concentration of antibiotics along with heavy metals and/or PAHs present in the environment, to estimate the contribution of these pollutants in the development and persistence of ARGs. The bacterial resistomes in the environment has not been given due considerations, for their potential role in ARGs evolution, and need to be studied aggressively to restrict the expansion of AMR. It is highly recommended that, the microbial ecosystems of contaminated soil, in correlation to selective pressure present, should be characterized, which is now much easier due to advances in sequencing technology and environmental genomics. Further, policies for the use of antibiotics, heavy metal, crude oils should

also be in place as a routine practice before discharging these pollutants into the environment and recognize the co-resistance and co-selection mechanisms before use in the different setting to reduce the chance of development of cross-resistance.

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